

Low-Temperature Chemical Synthesis and Microstructure Analysis of GeO₂ Crystals with α -Quartz Structure

Victor V. Atuchin,[†] Tatiana A. Gavrilova,[‡] Sergey A. Gromilov,[§] Vitalii G. Kostrovsky,^{||} Lev D. Pokrovsky,[†] Irina B. Troitskaia,[†] R. S. Vemuri,[⊥] G. Carbajal-Franco,[⊥] and C.V. Ramana^{*⊥}

Laboratory of Optical Materials and Structures and Laboratory of Nanodiagnostics and Nanolithography, Institute of Semiconductor Physics, and Laboratory of Crystal Chemistry, Institute of Inorganic Chemistry, Siberian Branch of the Russian Academy of Science, Novosibirsk 630090, Russia, Laboratory of Solid State Chemistry, Institute of Solid State Chemistry and Mechanochemistry, Siberian Branch of the Russian Academy of Science, Novosibirsk 630128, Russia, and Department of Metallurgical and Materials Engineering, University of Texas at El Paso, El Paso, Texas 79968

Received September 8, 2008; Revised Manuscript Received December 19, 2008

ABSTRACT: Germanium dioxide (GeO₂) crystals were prepared by precipitation from water solution of ammonium germanate with nitric acid at a relatively low-temperature (100 °C). The grown GeO₂ crystals were characterized to study their microstructure and chemical analysis using X-ray diffraction (XRD), scanning electron microscopy (SEM), transmission electron microscopy (TEM), infrared (IR) spectroscopy, and Raman spectroscopy techniques. The results indicate that the grown GeO₂ crystals exhibit α -quartz type crystal structure. The lattice parameters obtained from XRD were $a = 4.987(4)$ Å and $c = 5.652(5)$ Å. The crystals formed were submicrometer in size and exhibited uniform shape and diameter, which is about 500 nm. The characterization using microscopic and spectroscopic measurements indicate a high structural quality of GeO₂ crystals, in terms of control for morphology, single-phase, and chemical bonding and local structure, grown using the present approach.

1. Introduction

The growth, characterization, and utilization of metal-oxide crystals for application in modern electronic and optical devices continue to be an interesting topic for theoretical and experimental investigations. Germanium dioxide (GeO₂) exhibits many interesting physiochemical properties for applications in optical, electronic, and optoelectronic devices.^{1–5} GeO₂ is a photoluminescence and dielectric material. It exhibits high values of dielectric constant, refractive index, thermal stability, and mechanical strength. It has been proposed that GeO₂ has a photosensor capability, which is confirmed by optical spectroscopic methods such as Raman spectroscopy and electron spin resonance spectroscopy coupled with structural measurements.^{1–3} Because of these fascinating optical and electronic properties, GeO₂ has been considered as a promising material for optical waveguides and nanoconnections in optoelectronic communications.

Synthesis and optimization of a set conditions in any physical or chemical process to grow a particular phase of GeO₂ crystals is very important as this material exhibits several polymorphs.^{15–21} The tetragonal α -phase with rutile type structure ($P4_2/mnm$, PDF 35–729^{15,16}), in which Ge is 6-fold coordinated, is stable up to 1047 °C. The trigonal/rhombohedral β -phase with α -quartz type^{17–20} ($P3_121$, PDF 04–498^{17,18} and $P3_221$, PDF 36–1463^{19,20}), in which Ge is 4-fold coordinated, is stable above 1035 °C. Several high-pressure phases and glassy structures of GeO₂ were also reported in the literature.²¹ These polymorphs mainly differ in their chemical properties, especially solubility, densities, and

indexes of refraction, and hence, the ultimate device applications depend on the phase stability and microstructure of GeO₂ crystals. The GeO₂ with α -quartz type is a blue photoluminescence material with peak energies around 3.1 and 2.2 eV with a potential for use in nonlinear optics, luminescence, optoelectronics, and integrated optics.

A great attention has been paid to the preparation of GeO₂ nanostructures and submicrometer crystals, such as nanoparticles, nanofibers, and nanosheets, in recent years. GeO₂ crystals are usually produced by either physical evaporation or thermal oxidation methods. Crystallization through these techniques leads to the formation of one-dimensional structures.^{4–7} Attempts were made to obtain the nanofibers by electrospinning process.⁸ The GeO₂ whiskers were fabricated by ablating Ge target at $T = 820$ °C with a pulsed KrF excimer laser in an argon atmosphere.⁹ Chemical precipitation and hydrothermal methods were also employed to grow three-dimensional GeO₂ structures with sizes from 15 nm to several centimeters depending on synthesis conditions.^{10–13} Most recently, Kim et al. demonstrated the growth of uncommon cone-shaped structures of GeO₂ using a thermal evaporation process.¹⁴ In present study, the low-temperature ($T = 100 \pm 5$ °C) chemical synthesis to prepare β -phase GeO₂ with α -quartz type structure is suggested and production of highly quality crystals is demonstrated. The microstructural and morphological characteristics of the obtained nanocrystals have been studied in detail. The results obtained are presented and discussed in this paper.

2. Experimental Section

(A) Materials. The starting material were pure α -phase GeO₂ powder (Umicore Electro-optic Materials) with optical grade, aqueous ammonia (20.27 wt %), nitric acid (69.23 wt %), and distilled water.

(B) Synthesis. Two steps were carried out to obtain a GeO₂ precipitate. At first, 0.2 g of GeO₂ was added to 50 mL of distilled water and heated under magnetic stirring to a temperature of 90 °C. Then, 1.6 mL of aqueous ammonia was slowly added into the solution to form an ammonium germanate aqua solution with a pH 10 and

* To whom correspondence should be addressed. E-mail: rvchintalapalle@utep.edu.

[†] Laboratory of Optical Materials and Structures, Institute of Semiconductor Physics, Siberian Branch of the Russian Academy of Science.

[‡] Laboratory of Nanodiagnostics and Nanolithography, Institute of Semiconductor Physics, Siberian Branch of the Russian Academy of Science.

[§] Institute of Inorganic Chemistry, Siberian Branch of the Russian Academy of Science.

^{||} Institute of Solid State Chemistry and Mechanochemistry, Siberian Branch of the Russian Academy of Science.

[⊥] University of Texas at El Paso.

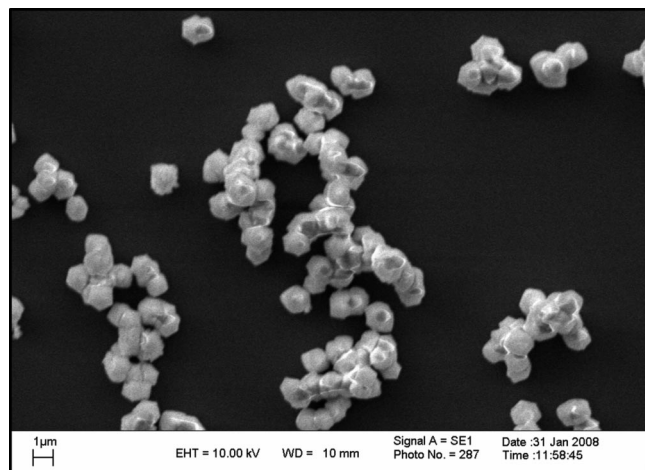


Figure 1. SEM image of as-prepared germanium dioxide deposit.

germanium concentration $C_{Ge} = 0.038$ mol/L. Under the conditions, the reaction between crystalline GeO_2 and aqueous ammonia produces germanate ions as monomers.²² At the second stage, 5 mL of nitric acid was added to the ammonium germanate solution and a white precipitate appeared, increasing the solution to 10 mL volume. The precipitate obtained has been separated by centrifugation and was washed with distilled water several times to pH 6 of washing water. The final product was dried in air at 80 °C.

(C) Characterization. The microstructure and morphology of GeO_2 nanocrystals was determined by scanning electron microscopy (SEM) using the LEO 1430 device. The accelerating voltage of the electron gun was 10 kV. Phase composition of the precipitate was examined by electron diffraction (ED) with EF-Z4 device with an operating accelerating voltage of 50 kV. A slow-electron gun was employed to account for charging effects (if any).

X-ray diffraction (XRD) pattern of the sample was recorded with DRON-UM1 device (Cu $K\alpha$ line, Ni filter). The scans were made from 5 to 90° 2θ .

Infrared (IR) spectra were recorded in the range of 500–2000 cm^{-1} using INFRALUM FT-801 (LUMEX) spectrometer with a spectral



Figure 2. SAED pattern for as-prepared β - GeO_2 with α -quartz type.

resolution of 1 cm^{-1} . Samples for IR measurements were prepared by pressing 1 mg of the material into a 500 mg, 1 mm thick KBr pellet. Raman spectra of the GeO_2 crystals were measured using RFS-100s spectrometer (Bruker) in the range of 20–1500 cm^{-1} at a spectral resolution of 1 cm^{-1} .

3. Results and Discussion

Surface morphology and distribution analysis of GeO_2 crystals performed using SEM indicate that the white hexagonal, rounded, or faceted monosized nanocrystals with typical diameters up to 500 nm were obtained. As it is evident from the SEM image shown in Figure 1, the morphology and dimension of the particles are uniform. Phase composition of the precipitate

Table 1. Interplanar Spacings (Å) in Germanium Dioxides

$P4_2/mnm$	$P3_121$	$P3_121$	$P3_221$	$P3_221$	$P3_221$
	4.32	4.3211	4.317	4.33	4.328
	3.429	3.4311	3.43	3.44	3.4268
3.11	2.496	2.4930	2.493	2.47	2.4864
2.399	2.366	2.3633	2.363		2.3472
	2.283	2.2807	2.280	2.25	2.267
2.199	2.159	2.1590	2.159		2.1554
2.106					
1.967	2.018	2.0170	2.016		2.0183
	1.884	1.8822	1.8827	1.87	1.8720
	1.87	1.8687	1.8687		
	1.726	1.7255	1.7257		1.7155
	1.716	1.7149	1.7150		
1.62	1.633	1.6321	1.6317		1.6282
1.5546	1.568	1.5678	1.5676		1.5658
	1.503	1.5024	1.5023		1.5012
1.4314	1.42	1.4184	1.4188		1.4092
	1.414	1.4127			
1.39	1.395	1.3943	1.3945	1.38	
	1.343	1.3417	1.3420		1.3405
1.3045					
1.3004					
	1.2830	1.2824	1.2822		1.2866
1.25	1.247	1.2462			
	1.234	1.2330	1.2331		1.2305
	1.231				
PDF 35–729	PDF 04–0497	PDF 36–1463	PDF 43–1016	TEM data ^a	XRD data ^a

^a Data are from this study.

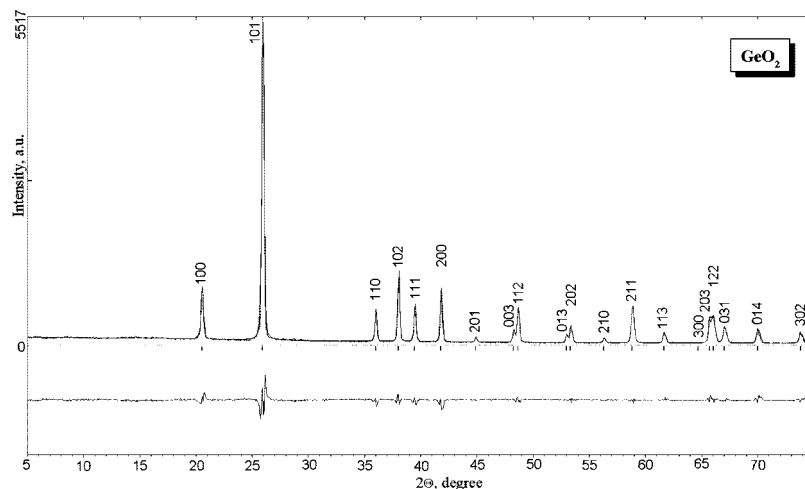


Figure 3. Observed (black line) and calculated (red line) X-ray diffraction pattern of as-prepared β -GeO₂ with α -quartz type. The difference reflection is shown at the bottom. Tick marks indicate allowed reflections.

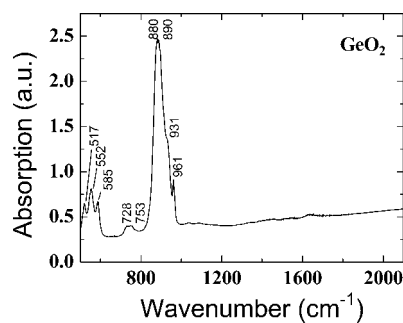


Figure 4. IR spectrum of β -modification of GeO₂.

as β -phase GeO₂ with α -quartz type ($P3_221$, PDF 43–1016) has been determined by TEM analysis. The presence of only β -phase of GeO₂ was found. Selected area electron diffraction (SAED) pattern is shown in Figure 2. The d -spacing values defined for the precipitate are listed in Table 1 together with those for other GeO₂ polymorphs and PDF 43–1016 card. It is evident from the comparison of the data presented in Table 1 that the structural data obtained in the present study is in close agreement with the data reported for β -phase GeO₂ with α -quartz type ($P3_221$, PDF 43–1016).

The powder X-ray diffraction pattern of the GeO₂ sample obtained is shown in Figure 3. XRD curve reveals good crystallinity and phase purity of as-prepared β -GeO₂. All the diffraction peaks were successfully indexed and their positions are in good agreement with those of the hexagonal β -GeO₂ with α -quartz type structure (space group $P3_221$) with lattice constants $a = 4.987(4)$ Å and $c = 5.652(5)$ Å. Comparison of d -spacing values obtained by the XRD analysis with the data obtained using TEM and data reported in the literature is presented in Table 1.

The IR spectrum recorded for as-prepared GeO₂ deposit is presented in Figure 4. The IR absorption bands and their respective positions observed are listed in Table 2. A comparison of the present IR data with that of literature data reported for various GeO₂ polymorphs is also presented in Table 2. The IR spectrum (Figure 4) and data (Table 2) demonstrates the excellent coincidence of spectral components with those of β -phase GeO₂.^{23,24} Strong absorption peak in the region of 885 cm⁻¹ is attributed to the vibration mode of GeO₄ tetrahedra.¹¹

Table 2. IR Absorption Bands (cm⁻¹) of GeO₂ Polymorphs

GeO ₂ (tetragonal)	GeO ₂ (rutile)	GeO ₂ (quartz)	GeO ₂ (trigonal)	β -GeO ₂ (trigonal)
		958	962	961
				931
		886	882	890
				880
				753
				728
715	700			
540	570			
		584		585
		554	585	552
		518	540	517
			491	
	435			
	347		345	
		332	332	
			265	
			249	
		212	209	
		124	122	
[29]	[27]	[27]	[30]	this study

However, in the present case, it is noticed that this IR band shows a noticeable splitting (at 880 and 890 cm⁻¹), which is not observed or reported in the literature for GeO₂. Such a splitting, which is an indicative of characteristic layered structure, was earlier observed in our work on oxides of Mo and V.^{31–33} Perhaps, this splitting observed could be due to the similar layered structure induced by distortion of GeO₄ tetrahedra in the GeO₂ nanocrystals. The characteristic sharp triplet at 517, 552, and 585 cm⁻¹ confirms a hexagonal structure of GeO₂ nanocrystals.¹¹ No band at 1620–1650 cm⁻¹, which is a characteristic of H₂O presence in the crystal, is observed in the IR spectrum recorded for the precipitate.

Raman spectrum measured for the GeO₂ nanocrystals is shown in Figure 5. The bands at 85, 125, 167, 214, and 262 cm⁻¹ correspond to the complex translation and rotation of the GeO₄ tetrahedra.²⁵ A shift in the band at 444 cm⁻¹ is due to symmetric Ge–O–Ge stretching. The presence of the bands at 516 and 592 cm⁻¹ is due to Ge–Ge stretching motions.²⁵ Bands at 860, 880, and 968 cm⁻¹ are assigned to Ge–O stretching motions with tetrahedral GeO₄ units. All the bands in the spectrum obtained for as-prepared sample are in good agreement with those of β -GeO₂ calcinated under $T = 1050$ °C.²⁶ Several new distinct bands are revealed in IR and Raman spectra,

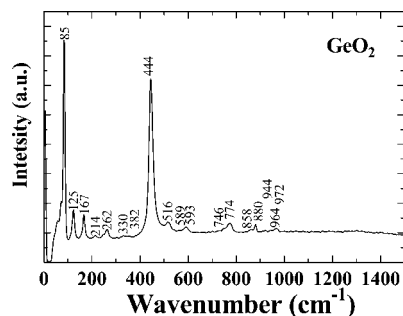


Figure 5. Raman spectrum of β -GeO₂ nanocrystals.

Table 3. Raman Lines (cm⁻¹) of GeO₂ Polymorphs

GeO ₂ (tetragonal)	GeO ₂ (rutile)	GeO ₂ (trigonal)	GeO ₂ (trigonal)	β -GeO ₂ (trigonal)
		972	973	972
		961	960	964
		949		944
873	870			
		880	881	880
		857	860	860
				774
				746
700	702			
	680			
		595	593	593
		583	583	589
		512	516	516
		492		
		456		
		440	444	444
		385		382
		326	330	330
		261	263	262
		212	213	214
170				
		166	166	167
		121	123	125
				85
[27]	97			
	[28]	[27]	[26]	this study

namely, 753 and 728 cm⁻¹ components in IR spectrum and 744 and 746 cm⁻¹ in Raman spectrum. Supposedly, these lines would be caused by scarce $P3_221$ symmetry obtained for GeO₂ nanocrystals synthesized under low temperature.

4. Conclusion

β -GeO₂ nanocrystals, which are traditionally formed at temperatures higher than 1050 °C, synthesized for the first time by a simplified wet chemical method at relatively low temperatures, as low as 100 °C. Characterization of the crystals using microscopic and spectroscopic methods reveals that the β -GeO₂ nanocrystals are single phase with a well-controlled morphology. The crystals are uniform in size and shape with obtained lattice parameters: $a = 4.987(4)$ Å and $c = 5.652(5)$ Å. The characteristic absorption bands observed in IR and Raman studies further confirms a hexagonal structure and excellent quality of GeO₂ nanocrystals without any presence of water.

Acknowledgment. C.V.R. acknowledges with pleasure the support of this work, in part, by the University of Texas at El Paso with the University Research Institute (URI) grant.

References

- (1) Terakado, N.; Tanaka, J. *Non-Cryst. Solids* **2006**, 352, 3815–3822.
- (2) Terakado, N.; Tanaka, K. *J. Non-Cryst. Solids* **2005**, 351, 54–60.
- (3) Zhou, M.; Shao, L.; Miao, L. *J. Phys. Chem. A* **2002**, 106, 6483–6486.
- (4) Su, Y.; Liang, X.; Li, S.; Chen, Y.; Zhou, Q.; Yin, S.; Meng, X.; Kong, M. *Mater. Lett.* **2008**, 62, 1010–1013.
- (5) Kim, H. W.; Lee, J. W. *Physica E* **2008**, 40, 2499–2503.
- (6) Jiang, Z.; Xie, T.; Wang, G. Z.; Yuan, X. Y.; Ye, C. H.; Cai, W. P.; Meng, G. W.; Li, G. H.; Zhang, L. D. *Mater. Lett.* **2005**, 59, 416–419.
- (7) Hidalgo, P.; M'endez, B.; Piqueras, J. *Nanotechnology* **2005**, 16, 2521–2524.
- (8) Kim, H. Y.; Viswanathamurthi, P.; Bhattarai, N.; Lee, D. R. *Rev. Adv. Mater. Sci.* **2003**, 5, 220–223.
- (9) Tang, Y. H.; Zhang, Y. F.; Wang, N.; Bello, I.; Lee, C. S.; Lee, S. T. *Appl. Phys. Lett.* **1999**, 74, 3824–3826.
- (10) Chen, X.; Cai, Q.; Zhang, J.; Chen, Z.; Wang, W.; Wu, Z.; Wu, Z. *Mater. Lett.* **2007**, 61, 535–537.
- (11) Jing, C.; Hou, J.; Zhang, Y. *J. Cryst. Growth* **2008**, 310, 391–396.
- (12) Balitsky, D. V.; Balitsky, V. S.; Pushcharovsky, D. Yu.; Bondarenko, G. V.; Kosenko, A. V. *J. Cryst. Growth* **1997**, 180, 212–219.
- (13) Wu, H. P.; Liu, J. F.; Ge, M. Y.; Niu, L.; Zeng, Y. W.; Wang, Y. W.; Lv, G. L.; Wang, L. N.; Zhang, G. Q.; Jiang, J. Z. *Chem. Mater.* **2006**, 18, 1817–1820.
- (14) Kim, H. W.; Shim, S. H.; Lee, J. W. *Appl. Surf. Sci.* **2007**, 253, 7207–7210.
- (15) Bolzan, A. A.; Fong, C.; Kennedy, B. J.; Howard, C. J. *Acta Crystallogr., Sect. B* **1997**, 53, 373–380.
- (16) Baur, W. H.; Khan, A. A. *Acta Crystallogr., Sect. B* **1971**, 27, 2133–2139.
- (17) Haines, J.; Cambon, O.; Philippot, E.; Chapon, L.; Hull, S. *J. Solid State Chem.* **2002**, 166, 434–441.
- (18) Glinemann, J.; King, H. E.; Schulz, H.; Hahn, T.; la Placa, S. J.; Dacol, F. Z. *Kristallogr.* **1992**, 198, 177–212.
- (19) Smith, G. S.; Isaacs, P. B. *Acta Crystallogr.* **1964**, 17, 842–846.
- (20) Yamanaka, T.; Ogata, K. *J. Appl. Crystallogr.* **1991**, 24, 111–118.
- (21) Prakapenka, V. P.; Shen, G.; Dubrovinsky, L. S.; Rivers, M. L.; Sutton, S. R. *J. Phys. Chem. Solids* **2004**, 65, 1537–1545.
- (22) Rimer, J. D.; Roth, D. D.; Vlachos, D. G.; Lobo, R. F. *Langmuir* **2007**, 23, 2784–2791.
- (23) Shvets, V.; Kas'yan, N. V.; Bezuglaya, E. M.; Tel'biz, G. M.; Il'in, V. G. *Theor. Exp. Chem.* **2001**, 37, 376–380.
- (24) Drugoveiko, O. P.; Evstrop'ev, K. K.; Kondrat'eva, B. S.; Petrov, Yu. A.; Shevyakov, A. M. *Zh. Prikl. Spektrosk.* **1975**, 22, 256–258.
- (25) Mernagh, T. P.; Liu, L.-g. *Phys. Chem. Miner.* **1997**, 24, 7–16.
- (26) Micoulaut, M.; Cormier, L.; Henderson, G. S. *J. Phys.: Condens. Matter.* **2006**, 18, R753–R784.
- (27) Scott, J. F. *Phys. Rev. B* **1970**, 1, 3488–3493.
- (28) Madon, M.; Gillet, Ph.; Julien, Ch.; Price, G. D. *Phys. Chem. Miner.* **1991**, 18, 7–18.
- (29) Kamitsos, E. I.; Yiannopoulos, Y. D.; Karakassides, M. A.; Chrysosikos, G. D.; Jain, H. *J. Phys. Chem.* **1996**, 100, 11755–11765.
- (30) Mirgorodsky, A. P. *Opt. Spectrosc.* **1973**, 34, 1146–1149.
- (31) Ramana, C. V.; Julien, C. M. *Chem. Phys. Lett.* **2006**, 428, 124–128.
- (32) Ramana, C. V.; Atuchin, V. V.; Pokrovosky, L. D.; Becker, U.; Julien, C. M. *J. Vac. Sci. Technol., A* **2007**, 25, 1166–1171.
- (33) Ramana, C. V.; Smith, R. J.; Hussain, O. M.; Chusuei, C. C.; Julien, C. M. *Chem. Mater.* **2005**, 17, 1213–1219.

CG8010037